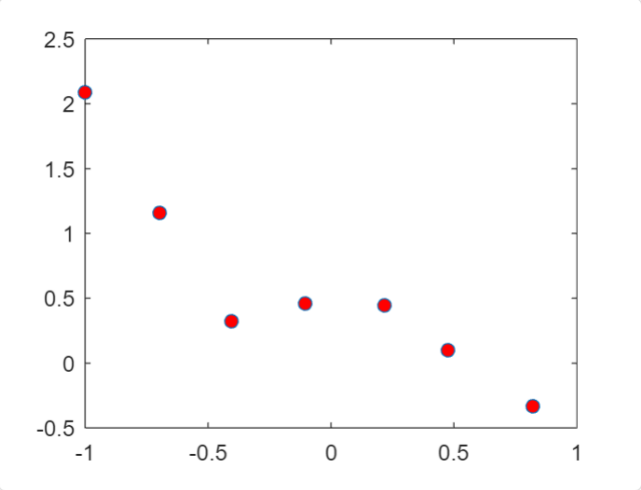
Experiment 6 : Regularization 1

1、插入数据，并绘制点图

```
load("ex5Linx.dat");
load("ex5Liny.dat");
```



2、将 x 变化为对应高次项，并添加截距项

```
for i = 1 : 7
    for j = 1 : 5
    X(i, j) = power(ex5Linx(i), j);
    end
end
x = [ones(7, 1), X];
```

3、定义特征矩阵 spe

```
spe = zeros(6, 6);
for  i = 2: 6
    spe(i, i) = 1;
end
```

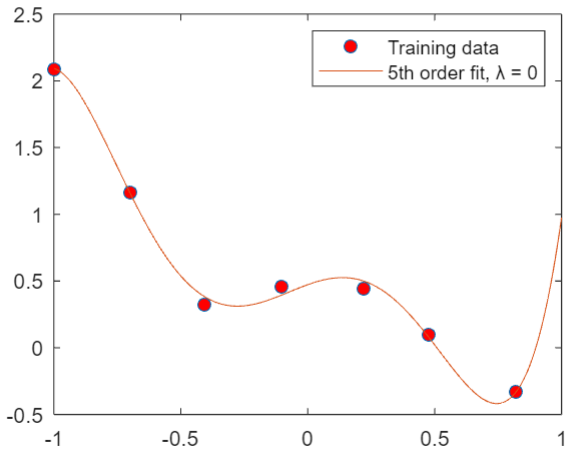
4、计算 θ 值(λ = 0，λ = 1，λ = 10)，并分别绘制图像

```
lamada = 10;
theta = inv((x'*x + lamada*spe))*x'*ex5Liny
plot(ex5Linx, ex5Liny, 'o','MarkerFaceColor', 'red');
hold on
plotx = linspace(-1,1,300);
for i = 1:300
```

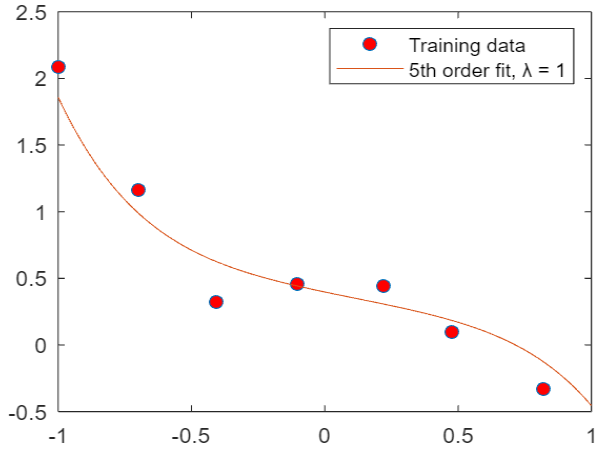
```
    ploty(i) = theta(1);
    for j = 2:6
    ploty(i) = ploty(i) + theta(j)*power(plotx(i), j-1);
    end
end
plot(plotx, ploty);
legend("Training data","5th order fit, λ = 10");
```

5、得到结果如下：

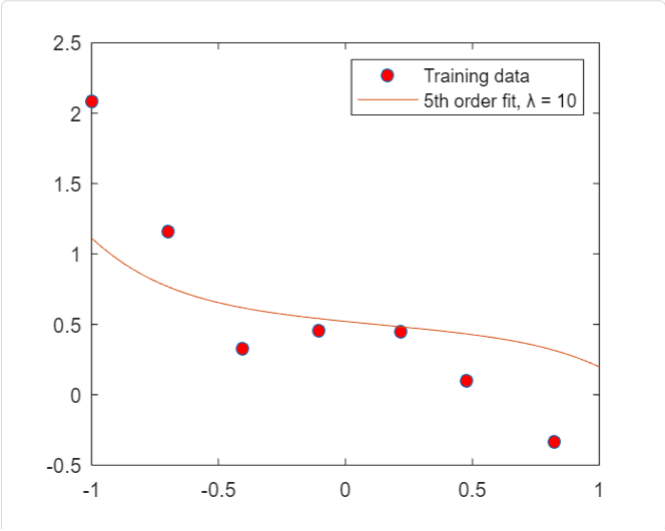
```
theta = 6×1
    0.4725
    0.6814
   -1.3801
   -5.9777
    2.4417
    4.7371
```



```
theta = 6×1
    0.3976
   -0.4207
    0.1296
   -0.3975
    0.1753
   -0.3394
```




```
theta = 6×1
    0.5205
   -0.1825
    0.0606
   -0.1482
    0.0743
   -0.1280
```

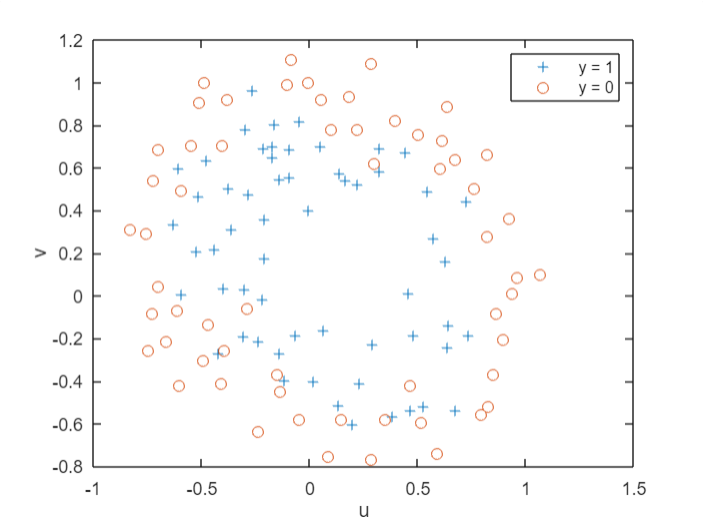


得出结论，λ值越大，得到的模型最高次幂越小，模型越简单

Experiment 6 : Regularization 2

1、插入数据，并绘制点图

```
x = load ("ex5Logx.dat") ;
y = load ("ex5Logy.dat") ;
figure
% Find t h e i n d i c e s f o r t h e 2 c l a s s e s
pos = find (y) ; neg = find (y == 0) ;
plot ( x(pos ,1) , x(pos,2),'+')
hold on
plot ( x(neg, 1) , x(neg,2),'o')
xlabel("u"); ylabel("v");
legend(" y = 1"," y = 0")
```



2、使用 2 维数据生成 28 维数据，并使用 28 维 theta，进行正则化牛顿法迭代求 theta 值

```
x2 = map_feature(x(:,1), x(:,2));
theta  = zeros(28,1);
m = 117; lamuda = 1;
n  = 28;
for i = 1 : 15
    theta = theta - inv(Hessian(m, lamuda, x2,theta))*J(lamuda, n, m ,theta,
x2, y);
end
```

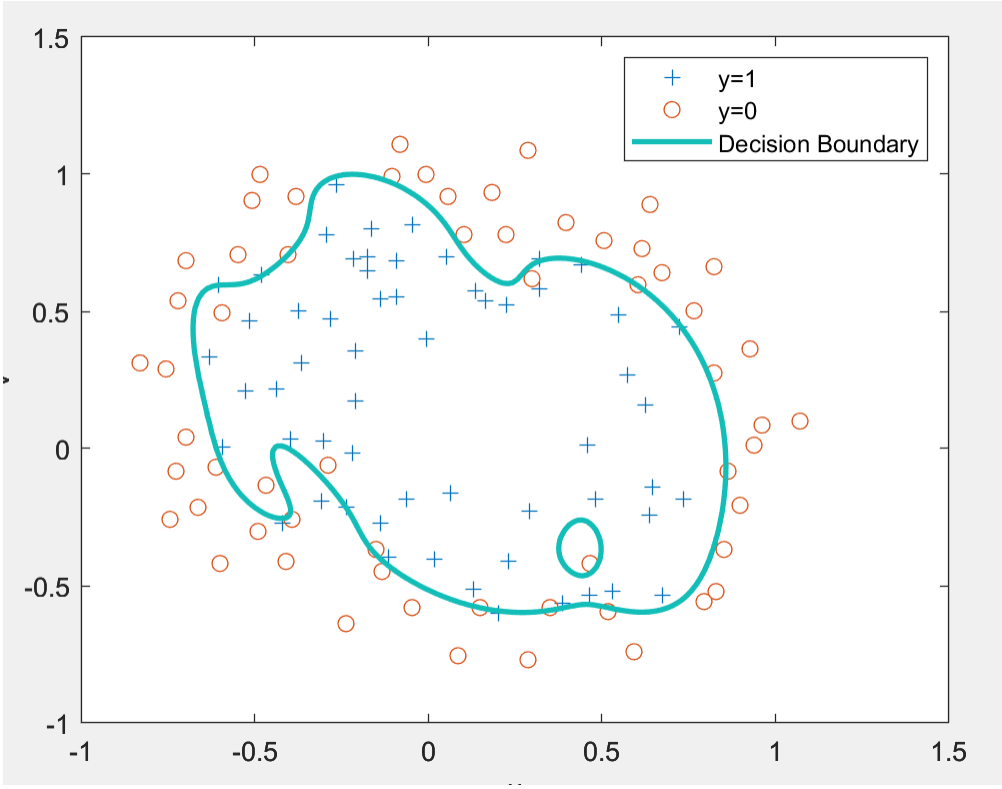
3、画图（λ = 0，λ = 1，λ = 10）

```
u = linspace(-1, 1.5, 200);
v = linspace(-1, 1.5, 200);
```

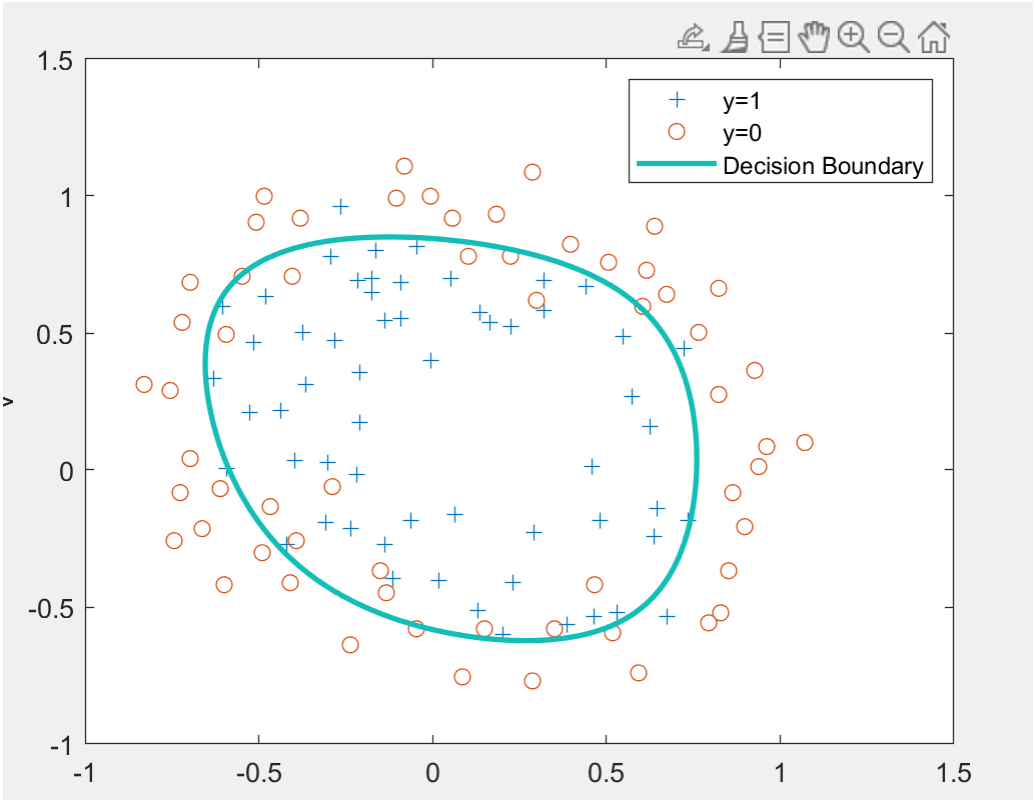
```
z = zeros(200, 200);
for i = 1 : length(u)
    for j = 1 : length(v)
        z(i,j) = map_feature(u(i), v(j))*theta;
    end
end
z  = z';
contour(u, v, z, [0, 0],'LineWidth',2);
legend('y=1','y=0','Decision Boundary');
```

得到结果如下:

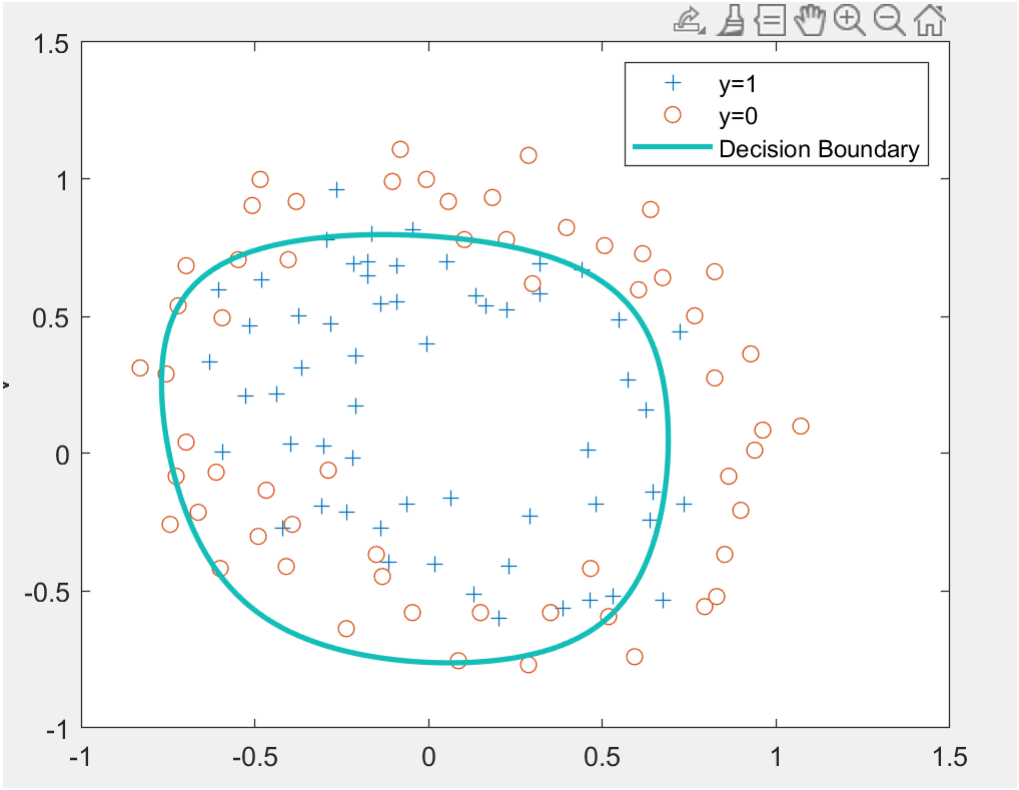
λ = 0



λ = 1


y=1
y=0
Decision Boundary
1.5
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
1.5

$\lambda = 10$


y=1
y=0
Decision Boundary
1.5
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
1.5

4、计算 L2 范数

```
L2 = 0;
for i = 1:28
L2  = L2 + theta(i)*theta(i)
end
L2 = L2*lamuda/(2*m);
```

λ = 0 时 L2 = 0， λ = 1，时 L2 = 0.0768， λ = 10 时，L2 = 0.8806

λ越大，L2 范数越大，曲线越圆滑

其他函数附录

%hx.m

```
function [g] = hx(Theta, ex4x)
g = 1.0/(1.0 + exp(-Theta'*ex4x));
end
```

%J.m

```
function [result] = J(lamuda, n, m ,theta, x, y)
    result = zeros(28,1);
    for j = 1 : n
        for i = 1 : m
            result(j) = result(j) + ((hx(theta, x(i, :)') - y(i))*x(i, j) / m);
        end
        if(j ~= 1)
            result(j) = result(j) + lamuda*theta(j)/m;
        end
    end
end
```

%Hessian.m

```
function [H] = Hessian(m, lamuda, x,theta)
    one_zeroMatrix = zeros(28, 28);
    H = zeros(28,28);
    for i = 2 : 28
        one_zeroMatrix(i, i) = 1;
    end
    for i = 1 : m
        H = H + hx(theta, x(i,:)')*(1 - hx(theta, x(i,:)'))*x(i,:)'*x(i,:);
    end
    H = H + lamuda*one_zeroMatrix;
    H = H/m;
end
```